DSL-BC: Deep Subspace Learning With Boundary Consistency for Hyperspectral Image Classification

Yun Cao, Yuebin Wang, *Member, IEEE*, Junhuan Peng, and Liqiang Zhang, *Member, IEEE*

***Abstract*— Deep subspace learning (DSL) plays an essential role in hyperspectral image (HSI) classification, providing an effective solution tool to reduce the redundant information of HSI pixels. Semisupervised convolutional neural network (SSCNN)-based DSL methods can extract a more representative representation of latent subspace with the help of the labeled and unlabeled data. However, CNN-based DSL methods may lose the information on the class boundaries leading to misclassifications within regular input. We develop the DSL method with boundary consistency (DSL-BC) for the HSI classification to address this problem. In DSL-BC, the convolutional autoencoder (CAE) is first applied to extract the deep subspace representation (DSR). The DSR is used to model the boundary consistency. The graph convolutional network (GCN) is further adapted to enforce the boundary consistency by conducting the graph convolution on arbitrarily structured non-Euclidean data and irregular image regions. In addition, the adaptive entropy rate (ER) superpixel segmentation algorithm is applied to generate superpixels, and superpixel constraint is employed to improve the ability of DSL and GCN construction. DSL-BC integrates the DSL, the GCN, and the superpixel constraint into a unified objective function. A customized iterative algorithm is used to solve the objective function of the DSL-BC. The experimental results on three challenging public HSI datasets demonstrate that the DSL-BC can outperform the related state-of-the-art HSI classification methods.**

***Index Terms*— Deep subspace learning (DSL), graph learning, hyperspectral image (HSI) classification, optimization, semisupervised learning.**

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I. Introduction

THE classification of the hyperspectral image (HSI) gives its category to the spectral of the pixel according to the spectral or spectral–spatial information [1]. This is important for various practical applications such as land-use mapping, mineral exploration, and urban resources management [2]–[4]. The high dimensionality of the HSI and the lack of labeled samples make the HSI classification challenging [5].

Subspace learning (SL) can provide an effective solution tool for reducing the redundant information of the HSI, which seeks a projection from the original high-dimensional to the low-dimensional subspace [6]. Therefore, SL plays an essential role in the HSI classification. With the development of convolutional neural networks (CNNs) and their applications to the SL, classification accuracy on HSIs has significantly been improved [7], [8]. The commonly used design in CNNs is based on stacked convolutions and pooling operations, which can learn the SL representation deeply [9]. Concerning the availability of labeled samples, the CNN-based SL methods can be split into three categories, that is, supervised, unsupervised, and semisupervised methods [10]. The supervised CNN-based SL methods obtain the classification results using labeled samples, which leads to more accurate classification maps than unsupervised approaches [11], [12]. However, the labeled samples may not be sufficient in real applications due to the limited time and labor [5]. The supervised CNN methods would be limited to a few labeled samples. Unsupervised CNN-based SL methods obtain classification results that do not consider the labeled samples [13], [14]. The autoencoder (AE) is an essential branch of the unsupervised CNN methods, extracting the latent subspace representation with the reconstruction error [15]–[17]. However, the results of the unsupervised CNN methods are limited with lacking the label information. The training step of the semisupervised CNN (SSCNN)-based SL methods is based on the unlabeled and labeled data [18]–[20]. With the guidance of the labeled samples, the extracted in-depth features from the unlabeled samples can be more representative and discriminative [21].

Although CNN-based SL methods benefit learning high-level feature representations, the geometric variations of diverse object regions of HSI cannot be captured adaptively. During the deep subspace representation (DSR) abstraction process, the CNN-based SL methods may lose the class boundaries due to the inflexible convolution kernel, leading to misclassifications [22]. The challenge is to preserve accurate boundaries. Graph convolutional network (GCN) operates on

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a graph, applying to the irregular regions with a predefined graph [23]. Therefore, the class boundaries can be flexibly preserved. Besides, with the neighbors of every graph node, GCN can aggregate and transform feature information to exploit the spatial–spectral features. However, there are two main HSI classification challenges directly using traditional GCN. The first challenge is that the original graph may be inaccurate due to the noise of the data. It usually contains much spectral–spatial information. The second challenge is that traditional GCN classifies HSIs in a pixel-based manner, which may generate the "salt-and-pepper" problem.

Superpixel represents a set of spectrally similar and spatially nearest pixels, while the object boundaries and the semantic structural information can be preserved to prevent the salt-and-pepper problem [24]. Pixels belonging to one superpixel are expected to have similar latent subspace representation and the same categories. However, the purity of each superpixel cannot be well ensured [25]–[27]. The performance will decrease if one pixel's label is different from that of other pixels within a superpixel. Considering the above-mentioned problems, the superpixel is introduced, and the label purity of the pixels within a superpixel is considered.

Both advantages can be considered for the HSI classification to solve the above-mentioned problems. This article proposes the DSL method with the boundary consistency (DSL-BC) for HSI classification. Fig. 1 illustrates the structure of our proposed approach. First, we adopt the convolutional AE (CAE) to obtain the compact DSR from the original HSI data with the reconstruction error. However, the CAE failed to adaptably catch the geometric variations of diverse object regions, which may lose the information on the class boundaries. The GCN can provide an excellent approach to keeping the boundary consistency, performing convolutions on the irregular patches with the predefined graph. The error of the GCN is measured using the cross-entropy (CE) loss (CEL). Moreover, an adaptive entropy rate (ER) method is employed to generate superpixels, and the DSR is further processed with the superpixel constraint to improve the ability of SL and GCN construction. The superpixel constraint can offer helpful feedback, and the salt-and-pepper problem can be avoided. The CEL is processed with the labeled samples, while the reconstruction loss and the superpixel constraint deal with all samples. The DSL-BC integrates the reconstruction loss, CEL, and superpixel constraint into the unified objective function, optimized with an iterative algorithm to obtain representative features. For clarity, we list the important notations and definitions used throughout this article in Table I.

The main contributions of this article are summarized as follows.

1) We propose the DSL-BC to capture the DSR and enforce the boundary consistency, containing the DSL and GCN. The CAE is adopted to obtain the DSR, and GCN can keep the boundary consistency by using the graph convolution.
2) The superpixel constraint provides valuable feedback for the DSL and GCN and avoids the salt-and-pepper problem. DSL-BC integrates the DSL, GCN, and superpixel constraint into the final objective function, formulating the end-to-end DSL framework.
3) DSL-BC is optimized using a customized algorithm. Extensive experiments on three public HSI datasets demonstrate the advantages of DSL-BC over the related state-of-the-art methods.

TABLE I
NOTATIONS AND DEFINITIONS OF DSL-BC

Notation	Definition
$\mathbf{X}$	HSI data. $\mathbf{X} \in \mathbb{R}^{b\times c\times d\times n}$.
$\mathbf{H}^{(l)}$	The output of l-th layer of the encoder in CAE. $1 \le l \le l_2$
$\mathbf{H}^{(l_2)}$	The deep subspace representation.
$\mathbf{H}^{(k)}$	The output of k-th layer of the decoder in CAE. $1 \le k \le k_2$
$\mathbf{H}^{(k_2)}$	The reconstructed HSI data.
θ_{CAE}	The weights and bias vectors included in the CAE.
θ_{GCN}	All parameters included in the GCN.
$\mathbf{A}$	The adjacency matrix. $\mathbf{A} \in \mathbb{R}^{n\times n}$
$\mathbf{Z}^{(m)}$	The output of m-th layer of GCN. $1 \le m \le M+1$
α_1, α_2	The learning rate of CAE and GCN, respectively.
λ_1, λ_2	Balance the corresponding terms.
d	The number of bands.
n	The number of samples.
P_i	The position of pixel i.

II. RELATED WORKS

A. Semisupervised Learning

The semisupervised learning methods can capture valuable information from a large number of unlabeled samples, which compensates for the lack of labeled data. The semisupervised learning methods for HSI classification can be divided into four categories: consistency regularization, proxy-label methods, generative models, and graph-based methods [28].

The consistency regularization methods tend to give consistent predictions for similar data. Liu *et al.* [29] proposed a novel semisupervised deep feature learning with the label consistencies model for HSI classification, where label consistencies of single pixels and group pixels are formulated. The proxy-label methods apply the prediction function to produce the proxy labels of unlabeled data in an unsupervised way. The proxy labels are trained with corresponding unlabeled data and the labeled data. In [19], the unlabeled data are trained with their pseudo-labels for HSI classification. The pseudo-labels are obtained by performing clustering on all training data. Variational AE and generative adversarial networks (GANs) are two typical generative models. He *et al.* [20] proposed the GAN-based semisupervised learning for HSI classification. GAN is trained on the spectral–spatial features by adding samples from the generator to the features. The graph-based methods construct interactions and relations between different entities. Recently, GCN is emerging and gaining increasing research interest, where the node features are used to generate the embeddings. More detailed information on GCN can be found in Section II-C.

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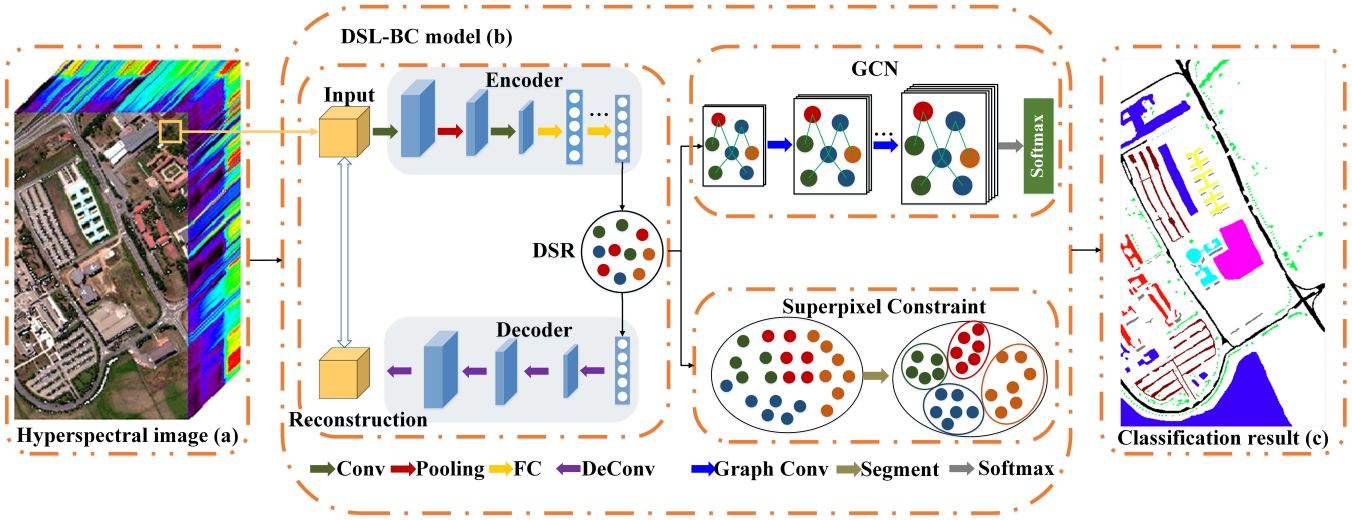


Fig. 1. Structure of the proposed DSL-BC method. (a) HSIs. (b) DSL-BC model. (c) Classification results. In (b), the DSR is first extracted from the CAE by the DSL, which consists of an encoder and a decoder. The circles with different colors denote the DSR with different classes. The GCN further constructs the DSR, in which the circles with different colors correspond to the different graph nodes. The green lines represent the graph edges. The final layer of GCN is activated with the softmax. In the superpixel constraint, the DSRs are segmented into several superpixels, where the DSRs with similar characteristics are clustered to a superpixel. The superpixel constraint enforces the DSR within a superpixel more compact.

B. Subspace Learning

SL refers to projecting the original image into points in the N-dimensional subspace to learn the features more effectively. There are two classic algorithms: principal component analysis (PCA) [30] and linear discriminant analysis (LDA) [31]. PCA aims to search for a proper subspace where variables have the most significant difference [30]. The features are extracted unsupervised for PCA, which may not discriminate in classification tasks. On the contrary, LDA learns the features in a supervised way. The features extracted by LDA are comparatively discriminative [31]. However, for LDA, the sample set needs to obey a Gaussian distribution. Moreover, the low-dimensional manifold methods assume that the inherent components of original data depend on the low-dimensional manifold, which has more discriminative power. Popular methods include locally embedded analysis [32], neighborhood preserving embedding [33], local discriminant embedding [34], locality preserving projection [35], and marginal Fisher analysis [36]. The above methods learn the subspace shallowly, whose abilities are limited.

Another branch of SL deeply learns features known as DSL, a deep transformation of feature extraction. The AE is a typical method of DSL to obtain latent subspace. In [16], a novel stacked sparse AE guided by latent relationship (LRSSAE) is proposed, where the latent relationship can be effectively exploited under the LRSSAE framework. In [37], the SL combined with the conditional random field (CRF) was proposed for the HSI classification. The subspace representation is extracted using the 3-D CAE. The CRF is established, where the pixel probabilities belonging to different labels and pixel relationships are used to construct the CRF unary and pairwise terms, respectively. Some DSL methods learn a latent shared subspace. Hong *et al.* [7] proposed the common DSL (CoSpace) framework by jointly combining DSL and supervised land cover classification, which can effectively link the learned features and label information. The CoSpace learns the shared latent subspace from the HSI-multispectral image. In [18], a deep semisupervised shared SL model is proposed for HSI classification. The primary feature extraction network extracts the features from high-dimensional space and fuses the features into one shared subspace. Then, the deep shared SL network can obtain a deeper and more representative subspace.

Based on the above studies, we propose the practical and profound DSL approach that introduces the superpixel constraint to learn the latent subspace more accurately.

C. Graph Convolutional Network

GCN has been successfully used in various tasks of computer vision, including semantic segmentation [38], image segmentation [23], action recognition [39], and time-series forecasting [40]. Kipf and Welling [23] proposed the semisupervised learning GCN method on graph-structured data for image classification. Zhang *et al.* [38] proposed the dual-graph CNN for semisupervised image classification, which performed graph convolution from different raw data views. Fu *et al.* [41] introduced Hessian GCN for semisupervised image classification, while the one-order spectral graph Hessian convolutions are optimized. Ghosh *et al.* [42] proposed a novel stacked spatio-temporal GCN for action segmentation. The stacked hourglass architecture was introduced to improve generalization performance and localization accuracy. Some attention has been paid to the GCN for the HSI classification. In [43], a semisupervised spectral–spatial GCN for HSI classification was proposed, where the proposed method explicitly applied the adjacency nodes in the graph to approximate the convolution. In [44], the minibatch GCN was proposed to optimize the large-scale GCNs in a minibatch strategy. The out-of-samples data can be inferred by minibatch GCN without retraining networks, which improves the classification results. In [29], the deep feature learning method is trained with label consistencies for the HSI classification.

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Label consistencies of a single pixel and group pixels were added to the features extracted by the GCN. In [45], GCN was employed to convolve arbitrarily structured non-Euclidean data suitable for irregular regions.

However, limited attention has been paid to the quality of the features extracted by GCN. This article adds the reconstruction error and superpixel constraint to the DSL-BC, improving the classification accuracy.

D. Superpixel-Based Constraints

A superpixel is a cluster of pixels that share similar features, whose shape and size can be adaptably adjusted for different spatial structures [24]. Superpixel is usually generated using graph-based algorithms and gradient-descent-based algorithms [26], [27]. The ER superpixel segmentation is one of the graph-based algorithms, which has been widely used for the HSI classification. A linear spectral clustering algorithm was proposed to segment superpixels, which can preserve the global properties of images [46]. Shen *et al.* [47] proposed a real-time image superpixel segmentation method, and the generated superpixels are equipped with high adherence to boundaries. A lazy random walk algorithm is proposed to segment superpixels, where the initial superpixels are iteratively optimized using the energy function.

The superpixel concept has been popular in the HSI classification recently. In [26], the subdictionary learning algorithm was introduced to the HSI classification under the sparse representation framework. The spatial context can be exploited with the adaptive multiscale superpixel strategy. Lu and Wei [27] proposed the multiscale superpixel-based active learning framework, introducing the multiscale superpixel maps to automatically label the selected informative samples. In general, the articles on using superpixel segmentation for the HSI classification introduce the multiscale superpixel classification strategy. In contrast, they do not exploit the adaptive superpixel segmentation, which segments the superpixels using the information learned from the previous iteration.

III. Deep Subspace Learning With Boundary Consistency

This section proposes a novel semisupervised method DSL-BC for the HSI classification. The motivation is first stated in Section III-A. Second, we present the procedure of the DSL in Section III-B. Third, the GCN is constructed in Section III-C, and the superpixel constraint is described in Section III-D. Next, the final objective function of the DSL-BC is formulated in Section III-E. Finally, the optimization of DSL-BC is provided in Section III-F.

A. Motivation

With the available unlabeled and limited labeled data, SSCNN-based SL methods capture more discriminative and representative latent subspace representation. However, the CNN-based SL methods may lose the information on the class boundaries due to their inflexible convolution kernel, leading to misclassifications. Through represented by graph topological information, the GCN's convolution operation can conduct on the structured non-Euclidean samples and irregular image regions. This article proposes the DSL-BC method to keep the boundary consistency using the GCN. In the DSL-BC, we first apply the CAE to extract the DSR of HSI. The GCN is further constructed by operating on the instance graphs to keep the boundary consistency. The feature information can be aggregated and transformed from the neighborhoods of each graph node. In addition, the adaptive ER superpixel segmentation algorithm is applied to generate superpixels. The superpixel constraint can provide useful feedback to the SL and GCN. DSL-BC integrates the DSL, the GCN, and the superpixel constraint into the final objective function. A customized algorithm is applied to update the parameters of the DSL-BC.

B. Deep Subspace Learning

Using the DSL, we can extract features from the unlabeled data to obtain the DSR. In this section, we describe the CAE to extract the DSR. The convolution and deconvolution operations of the CAE are adopted to describe the local space information and spectral relationships of the HSI data.

Given an HSI data $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n) \in \mathbb{R}^{b\times c\times d\times n}$, where $b \times c$ is the width and height of the HSI, d is the dimension of spectra, and n is the number of HSI samples. The HSI is fed into the CAE to obtain the DSR. The encoder of the CAE projects the HSI data into the DSR, which includes the l_1 convolutional layer, max-pooling layer, and $(l_2 - l_1 - 1)$ fully connected layers

$$\mathbf{H}^{(l)} = \begin{cases} \psi(\mathbf{W}^{(l)} \otimes \mathbf{H}^{(l-1)}) + \mathbf{b}^{(l)}, & 1 < l \le l_1 \\ P(\mathbf{H}^{(l_1)}) & \\ \psi(\mathbf{W}^{(l)} \cdot \mathbf{H}^{(l-1)}) + \mathbf{b}^{(l)}, & l_1 + 1 < l \le l_2 \end{cases} \tag{1}$$

where $\mathbf{H}^{(l_2)}$ is the DSR, and $1 < l_1 < l_2$. $\mathbf{H}^{(l)}$ is the output of the hidden layers in the encoder, where $1 \le l \le l_2$. When $l = 1$, we assume $\mathbf{H}^{(l-1)} = \mathbf{H}^{(0)}$, which is equivalent to the HSI data $\mathbf{X}$. ψ is the rectified liner unit (ReLU) function, and $P(\mathbf{H}^{(l_1)})$ is the max pooling operation to $\mathbf{H}^{(l_1)}$. $\otimes$ is the convolutional operation, and $\cdot$ denotes the fully connected operation. $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ $(1 \le l \le l_2)$ are weights and bias of the lth layer in the encoder, respectively.

The decoder reconstructs the HSI data from the DSR, which includes the k_1 deconvolution layers and $(k_2 - k_1)$ fully connected layers

$$\mathbf{H}^{(k)} = \begin{cases} \psi\left(\mathbf{W}^{(k)} \cdot \mathbf{H}_i^{(k-1)}\right) + \mathbf{b}^{(k)}, & 1 < k \le k_1 \\ \psi\left(\mathbf{W}^{(k)} \otimes (\mathbf{H}^{(k-1)})'\right) + \mathbf{b}^{(k)}, & k_1 < k \le k_2 \end{cases} \tag{2}$$

where $\mathbf{H}^{(k_2)} = (\mathbf{H}_1^{(k_2)}, \mathbf{H}_2^{(k_2)}, \ldots, \mathbf{H}_n^{(k_2)})$ is the reconstructed HSI data with the same size of $\mathbf{X}$. $\mathbf{H}^{(k)}$ is the output of the hidden layers in the decoder, where $1 \le k \le k_2$. We assume $\mathbf{H}^{(k-1)} = \mathbf{H}^{(0)}$ that is equivalent to the DSR, when $k = 1$. $\mathbf{H}^{(k-1)'}$ is the input of the deconvolution in the kth layer that have been padded with zeros. $\mathbf{W}^{(k)}$ and $\mathbf{b}^{(k)}$ $(1 \le k \le k_2)$ are weights and bias of the kth layer in the encoder, respectively.

The latent subspace can be acquired from the input HSI data through training the CAE with the reconstruction error.

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The objective function of the CAE is trained with training samples (both labeled and unlabeled), which can be formulated as follows:

$$\mathcal{L}_{\mathrm{CAE}}(\theta_{\mathrm{CAE}}) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{b\times c\times d}\left\|\mathbf{X}_i-\mathbf{H}_i^{(k_2)}\right\|_F^2+\frac{\mu}{2}\|\mathbf{W}\|_F^2$$
$$=\frac{1}{n}\sum_{i=1}^{n}\frac{1}{b\times c\times d}\left(\sum_{z=1}^{d}\sum_{y=1}^{c}\sum_{x=1}^{b}\left|(\mathbf{X}_i)_{x,y,z}-(\mathbf{H}_i^{(k_2)})_{x,y,z}\right|^2\right)+\frac{\mu}{2}\|\mathbf{W}\|_F^2 \quad (3)$$

where μ is the balancing parameter. $\mathbf{W}$ is the set of the weights of all the layers in CAE, θ_{CAE} concludes all the weights and bias vectors in the CAE. The first term is to formulate reconstruction error of the reconstructed HSI data $\mathbf{H}^{(k_2)}$ and the original HSI data $\mathbf{X}$ [17]. $(\mathbf{X}_i)_{x,y,z}$ represents the value at position (x, y, z) of the input $\mathbf{X}_i$, while $(\mathbf{H}_i^{(k_2)})_{x,y,z}$ represents the reconstructed value at position (x, y, z) of the reconstructed HSI data $\mathbf{H}_i^{(k_2)}$. The weight decay term $(1/2)\|\mathbf{W}\|_F^2$ can decrease the magnitude of the weights and avoid overfitting. With optimizing the objective function, the latent subspace of the corresponding HSI can be obtained.

The $\mathbf{H}^{(l_2)}$ that is obtained from the DSL is assigned to the DSR. We fed the DSR into the GCN to learn the latent relationship. In addition, the DSR is input into the superpixel constraint to calculate the distances between the training samples within a superpixel.

C. GCN Construction

Since the DSR has been obtained, the GCN learns the latent relationship among the DSR. As introduced in Section III-B, the DSR is extracted through the encoder of CAE. We embed the DSR into the GCN. The GCN is then operated on the instance graphs to construct the densely connected instance graphs. We treat each DSR as a graph node and build the relationships between the DSR using the edges in the graph. Let $\varsigma(\upsilon, \varepsilon)$ be the undirected graph. Let υ and ε be the collection of nodes and edges, respectively. The adjacency matrix $\mathbf{A}\in\mathbb{R}^{n\times n}$ indicates the connection information of every pair of nodes and can be defined using the following function:

$$\mathbf{A}=\begin{cases}\exp\left(-\dfrac{\left\|P_i-P_j\right\|_2^2}{\gamma}\right), & \mathbf{X}_i\in\mathcal{N}(\mathbf{X}_j)\text{ or }\mathbf{X}_j\in\mathcal{N}(\mathbf{X}_i)\\ 0, & \text{otherwise}\end{cases} \quad (4)$$

where γ is the alternative parameter. $\left\|P_i-P_j\right\|_2^2$ represents the spatial distance between pixel i and pixel j. $\mathcal{N}(\mathbf{X}_i)$ denotes the k_n-nearest neighborhood of the pixel $\mathbf{X}_j$.

Then, GCN is built by connecting several graph convolutional layers, where the nonlinear transformation of graph convolution can be formulated as

$$\mathbf{Z}^{(m)}=\varphi(\hat{\mathbf{D}}^{-\frac{1}{2}}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-\frac{1}{2}}\mathbf{Z}^{(m-1)}\mathbf{W}^{(m)}) \quad (5)$$

where $\hat{\mathbf{D}}^{-(1/2)}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-(1/2)}$ is the normalized adjacency matrix and describes the structure information. $\mathbf{Z}^{(m-1)}$ is the input of mth layer, while $\mathbf{Z}^{(m)}$ is the output of the mth layer, where $1\le m\le M$. We assume $\mathbf{Z}^{(m-1)}=\mathbf{Z}^{(0)}$ that is equivalent to the DSR $\mathbf{H}^{(l_2)}$, when $m=1$. $\hat{\mathbf{A}}=\mathbf{A}+\mathbf{I}$, where $\mathbf{I}$ is the identity matrix, and $\hat{\mathbf{D}}$ is the diagonal matrix with $\hat{\mathbf{D}}=\mathrm{diag}(\hat{d}_1,\hat{d}_2,\ldots,\hat{d}_n)$ and $\hat{d}_n=\sum_j\hat{\mathbf{A}}_{ij}$.

The CEL is calculated with the labeled data to measure the error of the GCN model. The GCN features of the final GCN layer $\mathbf{Z}^{(M)}$ are input to calculate the CEL that measures the error between the outputs of the DSL-BC and the true labels

$$\mathcal{L}_{\mathrm{GCN}}(\theta_{\mathrm{GCN}})=-\sum_{i=1}^{n_s}\sum_{j=1}^{C}I(j)\log\left(\mathbf{Z}_{ij}^{(M+1)}\right) \quad (6)$$

where n_s is the number of labeled nodes, and θ_{GCN} consists of all the parameters in the GCN. $I(j)$ equals 1 when j is the desired label of the ith sample ($1\le i\le n_s$); otherwise, the value is 0. The final output denotes the label prediction for all features

$$\mathbf{Z}^{(M+1)}=\sigma(\hat{\mathbf{D}}^{-\frac{1}{2}}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-\frac{1}{2}}\mathbf{Z}^{(M)}\mathbf{W}^{(M+1)})\in\mathbb{R}^{C\times n} \quad (7)$$

where C denotes the number of classes. σ is the softmax activation function.

D. Superpixel Constraint

The superpixel constraint is employed to prevent the salt-and-pepper problem and offer helpful feedback for DSL and GCN. Pixels within one superpixel with class purity are expected to agree with two assumptions.

1) Pixels within a superpixel should have homologous structures and labeling information.
2) The representation of the DSL within a superpixel should be consistent with each other.

To better represent the superpixel, we introduce the superpixel feature, and the average vector involved in the corresponding superpixel. We first adopt the ER segmentation to partition the entire image into n' nonoverlapping homogeneous superpixels. We regard each superpixel as an irregular region corresponding to a group of spatial neighboring and spectral similar pixels. Let $\mathbf{H}^{(l_2)}=(\mathbf{H}_1^{(l_2)},\mathbf{H}_2^{(l_2)},\ldots,\mathbf{H}_n^{(l_2)})\in\mathbb{R}^{V\times n}$ be the collection of n features in V dimension. Once the DSR $\mathbf{H}^{(l_2)}$ has been obtained, we further calculate the average vector involved in the corresponding superpixel as follows:

$$\mathbf{U}_i=\left(\sum_{i=1}^{q_i}\mathbf{H}_i^{(l_2)}\right)\Big/q_i \quad (8)$$

where q_i is the number of pixels within a superpixel. $\mathbf{U}_i$ represents the average vector of DSR within the superpixel, to which pixel i is located. We use $\mathbf{U}=(\mathbf{U}_1,\mathbf{U}_2,\ldots,\mathbf{U}_n)$ to represent the superpixel feature, where the pixels within a superpixel share the same superpixel feature.

The DSR $\mathbf{H}^{(l_2)}$ learned from the CAE is updated to obtain more accurate classification accuracy during the training process. It is inappropriate to segment superpixels from the original image to obtain the corresponding average vector of the DSR. Therefore, the adaptive superpixel segmentation method is adopted for the DSL-BC to generate superpixels using the DSR updated from the last iteration. The superpixel feature $\mathbf{U}$ is updated throughout the whole iterative process.

Since the DSR $\mathbf{H}^{(l_2)}$ and the superpixel feature $\mathbf{U}$ have been obtained, the superpixel constraint is optimized with all the labeled and unlabeled data to make the DSR involved in a

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superpixel more compact. The superpixel constraint can provide helpful feedback for the DSL and GCN construction and prevent the salt-and-pepper problem. The superpixel constraint can be measured by computing the squared Euclidean distance between the $\mathbf{H}_i^{(l_2)} \in \mathbb{R}^V$ and $\mathbf{U}_i \in \mathbb{R}^V$, which can be described as follows:

$$\mathcal{L}_{\mathrm{SC}} = \frac{1}{n}\sum_{i=1}^{n}\left\|\mathbf{H}_i^{(l_2)} - \mathbf{U}_i\right\|_2^2. \tag{9}$$

E. Objective Function Formulation of DSL-BC

DSL-BC is formulated as a semisupervised problem trained to simultaneously minimize the weighted sum of supervised and unsupervised loss functions. The CEL is trained in a supervised manner, while the reconstruction error and superpixel constraint are trained through the unsupervised way. Combining the constraints (3), (6), and (9), we formulate the objective function of DSL-BC as follows:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\mathrm{GCN}} + \lambda_1\mathcal{L}_{\mathrm{CAE}} + \lambda_2\mathcal{L}_{\mathrm{SC}} \\ &= -\sum_{i=1}^{n_s}\sum_{j=1}^{C} I(j)\log\left(\mathbf{Z}_{ij}^{(M+1)}\right) \\ &\quad +\lambda_1\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{b\times c\times d}\left\|\mathbf{X}_i - \mathbf{H}_i^{(k_2)}\right\|_F^2 + \frac{\mu}{2}\|\mathbf{W}\|_F^2\right) \\ &\quad +\lambda_2\frac{1}{n}\sum_{i=1}^{n}\left\|\mathbf{H}_i^{(l_2)} - \mathbf{U}_i\right\|_2^2\end{aligned} \tag{10}$$

where $\mathcal{L}_{\mathrm{GCN}}$ is the supervised loss, while $\mathcal{L}_{\mathrm{CAE}}$ and $\mathcal{L}_{\mathrm{SC}}$ are the unsupervised loss. λ_1 and λ_2 are the balancing parameters that are used to balance the related terms.

F. Optimization of DSL-BC

We summarize the customized iterative algorithm for DSL-BC in Algorithm 1. The objective function of DSL-BC is iteratively minimized. The reconstruction error, CEL, and superpixel constraint are optimized simultaneously using the Adam and back-propagation learning algorithm.

1) Optimization for θ_{CAE}: The parameters of CAE θ_{CAE} can be optimized with the following problem:

$$\begin{aligned}\mathcal{L}(\theta_{\mathrm{CAE}}) &= \mathcal{L}_{\mathrm{GCN}} + \lambda_1\mathcal{L}_{\mathrm{CAE}} + \lambda_2\mathcal{L}_{\mathrm{SC}} \\ &= -\sum_{i=1}^{n_s}\sum_{j=1}^{C} I(j)\log\left(\mathbf{Z}_{ij}^{(M+1)}\right) \\ &\quad +\lambda_1\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{b\times c\times d}\left\|\mathbf{X}_i - \mathbf{H}_i^{(k_2)}\right\|_F^2 + \frac{\mu}{2}\|\mathbf{W}\|_F^2\right) \\ &\quad +\lambda_2\frac{1}{n}\sum_{i=1}^{n}\left\|\mathbf{H}_i^{(l_2)} - \mathbf{U}_i\right\|_2^2.\end{aligned} \tag{11}$$

We calculate the derivative of (11) to the parameters θ_{CAE}, which is the collection of the weights and bias vectors in the CAE. Then, we update the parameters

$$\mathbf{W}^{(l)} = \mathbf{W}^{(l)} - \alpha_1\frac{\partial\mathcal{L}(\theta_{\mathrm{CAE}})}{\partial\mathbf{W}^{(l)}}, \quad \mathbf{b}^{(l)} = \mathbf{b}^{(l)} - \alpha_1\frac{\partial\mathcal{L}(\theta_{\mathrm{CAE}})}{\partial\mathbf{b}^{(l)}} \tag{12}$$

Algorithm 1 DSL-BC

Input:

Input: Training samples $\mathbf{X}$, balancing parameters: λ_1, λ_2. Initialize: θ_{CAE} and θ_{GCN}.

Output:

Predicted class labels.

for each epoch **do**
Sample batch size of samples $\mathbf{X}$;
Compute DSR from CAE using Eq. (1);
Compute $\mathbf{U}_i$ using Eq. (8);
Compute predicted classes using Eq. (5) and Eq. (7);
Update θ_{CAE} using Eq. (12) and Eq. (13);
Update θ_{GCN} using Eq. (15);

$$\mathbf{W}^{(k)} = \mathbf{W}^{(k)} - \alpha_1\frac{\partial\mathcal{L}(\theta_{\mathrm{CAE}})}{\partial\mathbf{W}^{(k)}}, \quad \mathbf{b}^{(k)} = \mathbf{b}^{(k)} - \alpha_1\frac{\partial\mathcal{L}(\theta_{\mathrm{CAE}})}{\partial\mathbf{b}^{(k)}} \tag{13}$$

where $1 \le l \le l_2$, $1 \le k \le k_2$. α_1 is the learning rate of CAE.

2) Optimization for θ_{GCN}: For θ_{GCN}, we rewrite the optimization problem defined in (10) as follows:

$$\mathcal{L}(\theta_{\mathrm{GCN}}) = \mathcal{L}_{\mathrm{GCN}} = -\sum_{i=1}^{n_s}\sum_{j=1}^{C} I(j)\log\left(\mathbf{Z}_{ij}^{(M+1)}\right). \tag{14}$$

We calculate the derivative of (14) to the parameters θ_{GCN}, where θ_{GCN} is the set of the weights of the GCN layers. Then, the parameters can be updated as follows:

$$\mathbf{W}^{(m)} = \mathbf{W}^{(m)} - \alpha_2\frac{\partial\mathcal{L}(\theta_{\mathrm{GCN}})}{\partial\mathbf{W}^{(m)}} \tag{15}$$

where $1 \le m \le M+1$. α_2 is the learning rate of GCN.

IV. Experiments

In this section, we use three simulated HSI datasets to evaluate the performance of DSL-BC. We first briefly describe the HSI datasets. Then, the classification results are compared.

A. Experimental Datasets

The experiments are processed on three widely used and changing datasets. We list the information of the three datasets in Tables II–IV.

The first dataset **Indian Pines** was gathered by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor. The dataset was acquired over the Indian pines area in 1992, consisting of 145×145 pixels with 20-m spatial resolution. The 224 spectral bands range from 0.4 to 2.5 μm. We abandon 20 spectral bands (104–108, 150–163, and 220), and preserve 200 bands. The 10 249 labeled pixels of the Indian pines dataset were classified into 16 classes.

The **Salinas** dataset was obtained from the 224-band AVIRIS sensor with 512×217 pixels. The spatial resolution is approximately 3.7 m. We abandon 20 spectral bands. The 54 129 labeled pixels of the Salinas dataset were classified into 16 classes.

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TABLE II
INDIAN PINES DATASET

Class	Land Cover Type	Labeling (5%)	Total
1	Alfalfa	2	46
2	Corn-notill	71	1,428
3	Corn-mintill	42	830
4	Corn	12	237
5	Grass-pasture	24	483
6	Grass-trees	36	730
7	Grass-pasture-mowed	1	28
8	Hay-windrowed	24	478
9	Oats	1	20
10	Soybean-notill	49	972
11	Soybean-mintill	123	2,455
12	Soybean-clean	29	593
13	Wheat	10	205
14	Woods	63	1,265
15	Bldg-grass-trees	20	386
16	Stone-Steel-Towers	5	93
	Total	512	10,249

TABLE III
SALINAS DATASET

Class	Land Cover Type	Labeling (1%)	Total
1	Weeds-1	20	1,989
2	Weeds-2	37	3,706
3	Fallow	19	1,956
4	Fallow-plow	14	1,374
5	Fallow-smooth	26	2,658
6	Stubble	39	3,939
7	Celery	35	3,559
8	Grapes	112	11,251
9	Soil	62	6,183
10	Corn	32	3,258
11	Lettuce-4wk	10	1,048
12	Lettuce-5wk	19	1,907
13	Lettuce-6wk	9	896
14	Lettuce-7wk	10	1,050
15	Vineyard-untrained	72	7,248
16	Vineyard-trellis	18	1,787
	Total	534	53,809

TABLE IV
PAVIAU DATASET

Class	Land Cover Type	Labeling (1%)	Total
1	Asphalt	66	6,631
2	Meadows	186	18,649
3	Gravel1	20	2,099
4	Trees	30	3,064
5	Metal sheets	13	1,345
6	Bare Soil	50	5,029
7	Bitumen	13	1,330
8	Bricks	36	3,682
9	Shadows	94	947
	Total	508	42,776

The **PaviaU** dataset is collected by the ROSIS-03 sensor over Pavia, northern Italy, consisting of 610×340 pixels. The 115 bands range from 0.43 to 0.86 μm. The spatial resolution is 1.3 m. We abandoned 12 spectral bands and preserved 103 bands. The PaviaU dataset contains nine categories and 42 776 labeled pixels.

B. Related Approaches

We adopt the following methods to compare with the proposed DSL-BC.

1) Active labeling method for deep learning (ALDL) [48]: ALDL is a semisupervised active learning approach. We select the entropy sampling for labeled data selection.
2) PCA + CE (PCA+CE): PCA is employed to project the HSI into the subspace. The feature in the subspace and its neighborhood are taken as the 3-D input patch. We feed the 3-D patches into the CNN and calculate the CEL with labeled samples.
3) Wasserstein GAN + CE (WGAN+CE) [49]: The WGAN optimizes with the adversarial loss and obtains the spectral–spatial features. Then, the labeled samples are trained with the CEL.
4) SSCNN [50]: The semisupervised SSCNN network learns the spectral–spatial features automatically from HSI. The SSCNN adds the skip connection parameters between the encoder and the decoder. The objective function is the weighted sum of supervised and unsupervised loss.
5) SESEMI [51]: The SESEMI method optimizes with the sum of supervised and self-supervised loss. The self-supervised loss is trained with the proxy labels, while the supervised CEL is trained with real labels.
6) SpectralFormer [52]: SpectralFormer is a new transformer-based backbone network. The groupwise spectral embedding (GSE) and cross-layer adaptive fusion are adopted to learn locally detailed spectral representations.
7) Semi-supervised deep feature learning with feature consistency (SDFL-FC) [53]: SDFL-FC is a semisupervised deep feature learning HSI classification method, enforcing the feature consistency of single pixels and group pixels.

In the compared methods and our proposed method, we randomly selected {1%, 2%, 3%, 4%, 5%} of each class as labeled samples for the Salinas and PaviaU datasets and {5%, 6%, 7%, 8%, 9%, 10%} for each class for the Indian pines dataset. The remaining part is test data. The number of feature dimensions in each layer is selected from {32, 64, 128, 256, 512}. The number of batch size is selected from {16, 32, 64, 128, 256, 512}, whereas the learning rate is selected from {0.00002, 0.00005, 0.0001, 0.001, 0.01}. The input sizes of HSI dataset are selected from {3, 5, 7}. The number of nonoverlapping superpixel regions is selected from {100, 300, 500, 700, 900, 1100}. The balancing parameters λ_1, λ_2 are selected from $\{0.01, 0.1, 1, 10, 100\}$. The alternative parameter is selected from {5, 10, 50, 100, 500, 1000, 5000, 10 000}.

C. Evaluation Results

In Tables V–VII and Figs. 2–5, we evaluate the performances of compared method and DSL-BC using the overall

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TABLE V
SEMISUPERVISED CLASSIFICATION RESULTS (%) WITH 5% LABELED SAMPLES FOR THE INDIAN PINES DATASET

Class	ALDL	PCA+CE	WGAN+CE	SSCNN	SESEMI	SpectralFormer	SDFL-FC	DSL-BC
Alfalfa	39.53	36.43	40.00	32.25	81.95	30.11	93.02	96.95
Corn-notill	90.47	84.99	74.83	87.04	88.90	77.42	97.61	97.86
Corn-mintill	83.21	71.79	67.00	83.38	94.76	71.44	93.09	99.90
Corn	81.40	58.28	52.74	53.93	92.46	47.35	90.53	96.79
Grass-pasture	86.93	92.10	77.29	90.36	97.07	81.89	96.81	99.29
Grass-trees	90.19	98.78	86.36	99.38	97.63	88.92	98.72	100.00
Grass-pasture-mowed	56.04	78.97	25.38	65.38	94.93	4.17	85.21	99.11
Hay-windrowed	96.32	100.00	95.10	99.16	98.96	96.15	99.61	99.48
Oats	26.32	85.96	48.42	38.95	0.00	13.15	86.64	91.90
Soybean-notill	79.35	69.02	65.98	75.44	86.03	75.07	94.27	99.56
Soybean-mintill	90.00	85.55	82.28	89.73	92.39	85.69	98.03	98.97
Soybean-clean	70.13	64.97	49.30	54.70	91.53	58.20	97.32	99.18
Wheat	98.53	100.00	99.90	99.90	99.89	93.46	99.84	98.73
Woods	97.64	97.60	94.79	98.00	94.75	92.49	99.85	100.00
Bldg-grass-trees	83.88	72.81	41.49	74.48	83.00	50.03	89.70	99.96
Stone-Steel-Towers	100.00	95.53	92.35	94.54	92.05	66.71	96.68	99.91
OA	87.86	84.04	76.57	86.06	92.21	79.30	96.92	**99.17**
	± 0.37	± 0.34	± 0.52	± 0.36	± 0.42	± 1.50	± 0.27	± 0.11
AA	79.37	80.80	68.33	77.29	86.64	64.51	94.81	**98.60**
	± 0.64	± 1.54	± 1.43	± 1.57	± 0.68	± 3.21	± 0.39	± 0.37
Kappa	0.861	0.817	0.733	0.840	0.911	0.763	0.965	**0.991**
	± 0.004	± 0.004	± 0.006	± 0.004	± 0.005	± 0.017	± 0.003	±0.001

The best results are highlighted in bold.

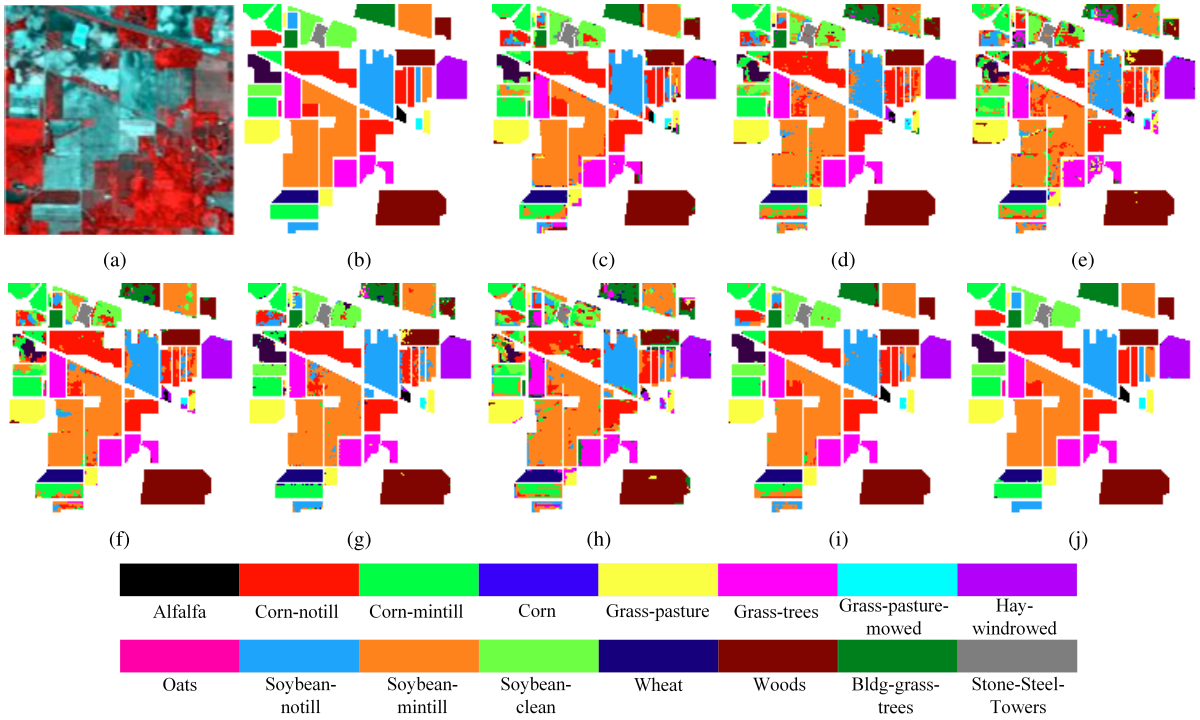


Fig. 2. Classification maps of the different methods with 5% labeled samples for the Indian pines dataset. (a) False-color. (b) Ground truth. (c) ALDL. (d) PCA+CE. (e) WGAN+CE. (f) SSCNN. (g) SESEMI. (h) SpectralFormer. (i) SDFL-FC. (j) DSL-BC.

accuracy (OA), the average accuracy (AA), and the Kappa coefficient. The results are reported over 15 times on the test samples, and we calculate the average and standard deviation of the results. We can further derive the following conclusions.

1) Compared with the related methods, the DSL-BC has the optimal results on testing data. For the three datasets, the DSL-BC classification accuracies are superior to these compared methods. It shows that the DSL-BC is practical and appropriate for the HSI classification.
2) For the three datasets, the HSI maps of DSL-BC are more compact in Figs. 2–4. This is because DSL-BC can build relationships and graphs between pixels. It also proves that the reconstruction error and superpixel constraint can give helpful feedback to improve HSI classification performance.

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TABLE VI
SEMISUPERVISED CLASSIFICATION RESULTS (%) WITH 1% LABELED SAMPLES FOR THE SALINAS DATASET

Class	ALDL	PCA+CE	WGAN+CE	SSCNN	SESEMI	SpectralFormer	SDFL-FC	DSL-BC
Weeds-1	99.06	99.20	98.29	98.39	99.60	94.94	99.96	100.00
Weeds-2	92.26	94.50	98.86	94.98	100.00	98.40	100.00	99.76
Fallow	86.79	92.69	89.16	92.74	89.22	93.24	99.92	99.91
Fallow-plow	88.97	88.77	99.78	82.68	99.78	97.7	96.71	99.18
Fallow-smooth	90.25	99.21	97.13	97.62	99.40	95.7	92.75	99.78
Stubble	99.23	99.67	100.00	99.49	100.00	99.67	100.00	99.99
Celery	98.65	97.43	98.59	97.69	99.94	97.61	99.11	99.95
Grapes	80.69	84.39	75.94	83.26	94.97	84.65	91.73	96.51
Soil	95.52	97.85	98.68	98.86	99.92	99.03	100.00	99.99
Corn	84.30	91.99	89.34	92.57	98.32	90.46	95.25	98.65
Lettuce-4wk	79.68	93.66	77.58	95.74	90.82	88.45	98.97	98.40
Lettuce-5wk	86.67	96.43	98.11	93.97	100.00	98.81	99.11	99.99
Lettuce-6wk	79.20	93.71	99.67	91.28	100.00	98.65	98.04	100.00
Lettuce-7wk	89.47	94.43	91.41	92.63	99.81	97.21	97.90	98.98
Vinyard-untrained	71.84	73.27	61.31	66.81	60.58	79.11	89.48	97.85
Vinyard-trellis	85.61	98.83	95.75	97.26	98.34	91.29	97.08	99.25
OA	86.85	90.72	87.39	89.37	92.86	91.67	95.85	**98.78**
	± 0.55	± 0.19	± 0.62	± 0.24	± 0.51	± 0.50	± 0.26	± 0.26
AA	88.01	93.50	91.85	92.25	95.67	94.07	97.23	**99.27**
	± 0.64	± 0.23	± 0.42	± 0.45	± 0.35	± 0.54	± 0.18	± 0.19
Kappa	0.854	0.897	0.860	0.882	0.920	0.907	0.954	**0.986**
	± 0.006	± 0.002	± 0.007	± 0.003	± 0.006	± 0.005	± 0.003	± 0.003

The best results are highlighted in bold.

TABLE VII
SEMISUPERVISED CLASSIFICATION RESULTS (%) WITH 1% LABELED SAMPLES FOR THE PAVIAU DATASET

Class	ALDL	PCA+CE	WGAN+CE	SSCNN	SESEMI	SpectralFormer	SDFL-FC	DSL-BC
Asphalt	90.96	93.39	89.92	95.68	86.57	84.60	96.61	97.68
Meadows	98.76	99.00	98.14	99.69	99.49	95.89	99.72	99.86
Gravel	78.89	82.51	77.77	74.60	69.37	59.46	90.85	96.10
Trees	90.20	86.91	85.69	88.54	95.21	91.30	93.51	96.65
Metal sheets	95.31	93.57	93.07	93.35	98.80	98.67	99.29	98.88
Bare Soil	74.40	86.53	81.19	93.34	49.82	71.59	96.39	98.92
Bitumen	86.37	80.13	78.15	77.05	92.73	66.28	93.48	98.69
Bricks	83.47	86.34	81.26	84.37	92.88	81.49	92.40	96.11
Shadows	99.54	96.65	95.33	93.31	99.51	94.10	99.63	92.16
OA	91.31	93.09	90.68	93.93	89.06	86.86	97.13	**98.44**
	± 0.15	± 0.12	± 0.51	± 0.22	± 0.45	± 0.87	± 0.20	± 0.09
AA	88.66	89.45	86.73	88.88	87.15	82.48	95.76	**97.23**
	± 0.26	± 0.19	± 0.69	± 0.39	± 0.74	± 0.85	± 0.31	± 0.14
Kappa	0.884	0.908	0.876	0.919	0.852	0.824	0.962	**0.979**
	± 0.002	± 0.002	± 0.007	± 0.003	± 0.006	± 0.012	± 0.003	± 0.001

The best results are highlighted in bold.

3) With increasing amount of labeled data, the accuracies are raised. DSL-BC achieves the higher OAs with varying the amount of labeled data. It demonstrates that the DSL-BC is superior to the compared methods. It is because the DSL-BC integrates the reconstruction error, GCN, and superpixel constraint into the final unified objective function. The reconstruction error removes the redundant information of spectra, while GCN can be applied to the irregular regions. Moreover, the superpixel constraint can provide helpful feedback to the DSL and GCN and prevent the salt-and-pepper problem. Therefore, DSL-BC has much better performance.

D. Independent Analysis of the Regularization Terms

To check each term's function, we compare the independent regularization (IR) and joint regularizations and divide the joint function into the following equations.

1) IR1: IR1 is the DSL, where the redundant information is removed using the reconstruction error. The CEL is trained with the DSR to provide the HSI classification results. The parameters θ_{CAE} can be optimized by calculating the derivative of $\mathcal{L}_{\mathrm{IR1}}$ to θ_{CAE}

$$\mathcal{L}_{\mathrm{IR1}} = \min_{\theta_{\mathrm{CAE}}} -\sum_{i=1}^{n_s}\sum_{j=1}^{C} I(j)\log\left(\mathbf{H}_{ij}^{(l_2)}\right) + \lambda_1\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{b\times c\times d}\left\|\mathbf{X}_i-\mathbf{H}_i^{(k_2)}\right\|_F^2+\frac{\mu}{2}\|\mathbf{W}\|_F^2\right). \tag{16}$$

2) IR2: To make the DSR more representative and discriminative, IR2 adds the constraints of GCN to the objective function based on IR1, where the CEL is connected to the output layer of GCN to measure the error of the GCN model.

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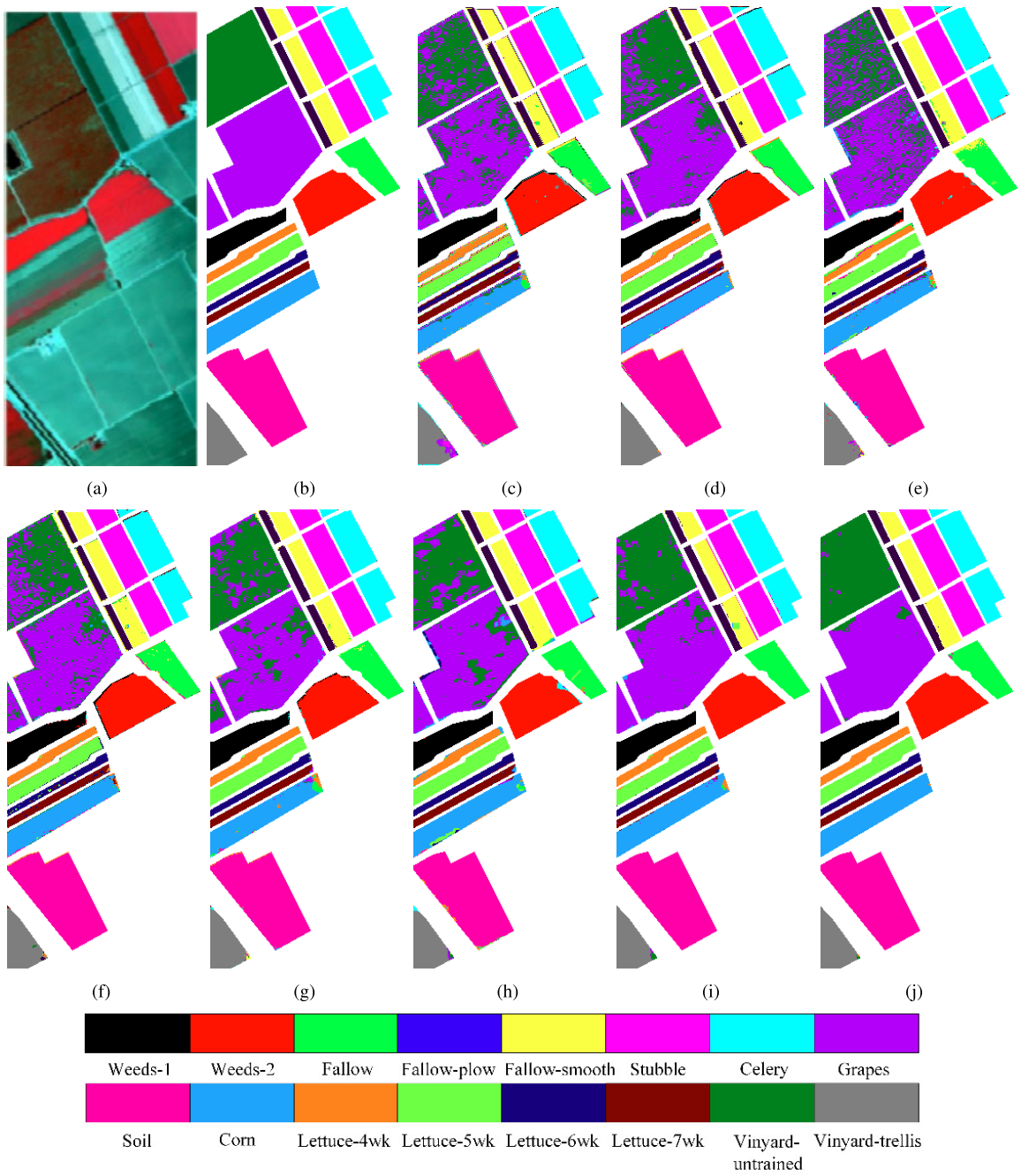


Fig. 3. Classification maps of the different methods with 1% labeled samples for the Salinas dataset. (a) False-color. (b) Ground truth. (c) ALDL. (d) PCA+CE. (e) WGAN+CE. (f) SSCNN. (g) SESEMI. (h) SpectralFormer. (i) SDFL-FC. (j) DSL-BC.

The parameters θ_{CAE} and θ_{GCN} can be optimized by calculating the derivative of $\mathcal{L}_{\mathrm{IR2}}$ to θ_{CAE} and θ_{GCN}

$$\mathcal{L}_{\mathrm{IR2}} = \min_{\theta_{\mathrm{CAE}},\theta_{\mathrm{GCN}}} -\sum_{i=1}^{n_s}\sum_{j=1}^{C} I(j)\log\left(\mathbf{Z}_{ij}^{(M+1)}\right) + \lambda_1\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{b\times c\times d}\left\|\mathbf{X}_i-\mathbf{H}_i^{(k_2)}\right\|_F^2+\frac{\mu}{2}\|\mathbf{W}\|_F^2\right). \tag{17}$$

3) IR3: Based on GCN, the superpixel constraint is added to provide helpful feedback and prevent the salt-and-pepper problem

$$\mathcal{L}_{\mathrm{IR3}} = \min_{\theta_{\mathrm{GCN}},\theta_{\mathrm{E\text{-}CAE}}} -\sum_{i=1}^{n_s}\sum_{j=1}^{C} I(j)\log\left(\mathbf{Z}_{ij}^{(M+1)}\right) + \lambda_2\frac{1}{n}\sum_{i=1}^{n}\left\|\mathbf{H}_i^{(l_2)}-\mathbf{U}_i\right\|_2^2. \tag{18}$$

where $\theta_{\mathrm{E\text{-}CAE}}$ is the parameters of encoder of CAE. The parameters $\theta_{\mathrm{E\text{-}CAE}}$ and θ_{GCN} can be optimized by calculating the derivative of $\mathcal{L}_{\mathrm{IR3}}$ to $\theta_{\mathrm{E\text{-}CAE}}$ and θ_{GCN}.

For the three datasets, s% labeled data are randomly selected (for Indian pines, $s = 5$; for Salinas and PaviaU, $s = 1$). We can drive the following observations from Table VIII and Fig. 6.

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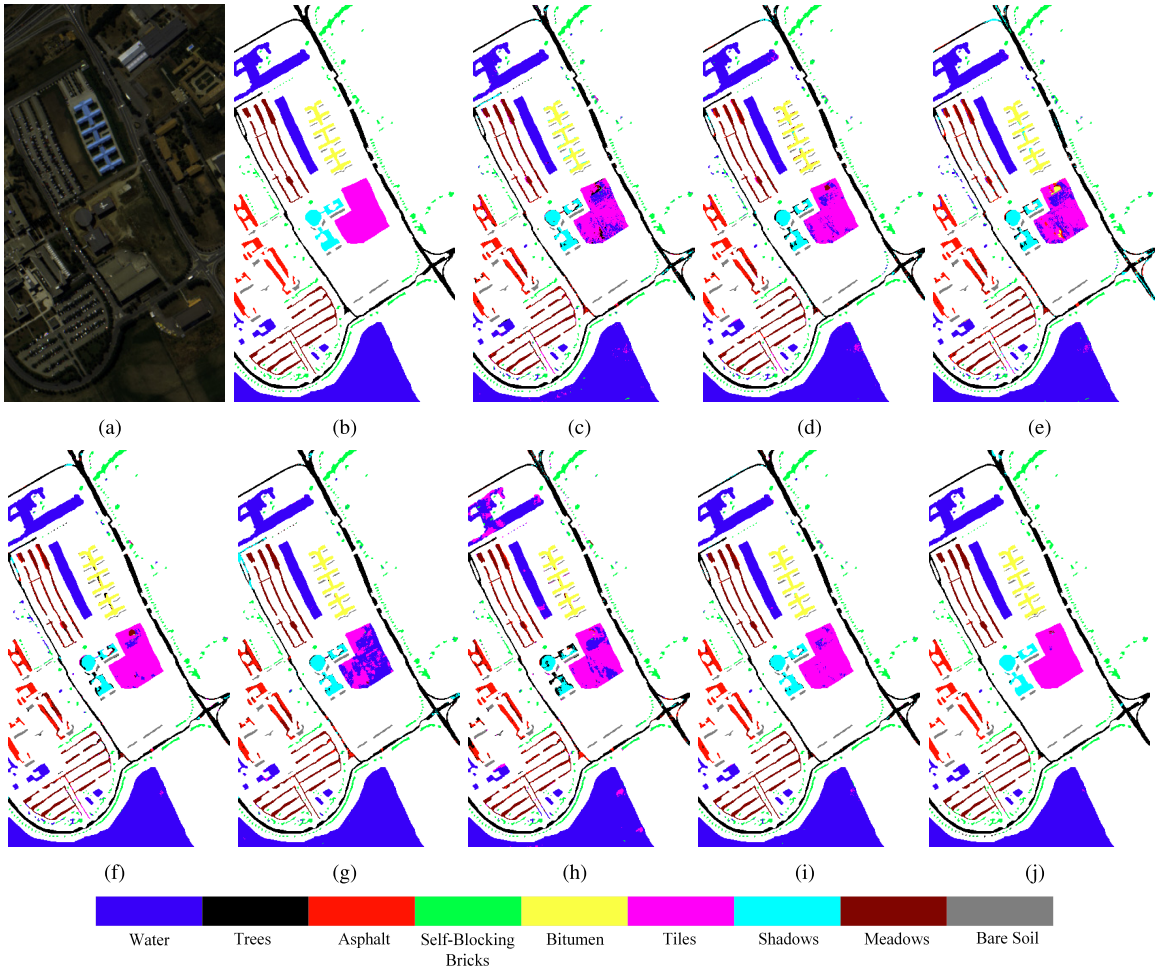


Fig. 4. Classification maps of the different methods with 1% percentage labeled samples for the PaviaU dataset. (a) False-color. (b) Ground truth. (c) ALDL. (d) PCA+CE. (e) WGAN+CE. (f) SSCNN. (g) SESEMI. (h) SpectralFormer. (i) SDFL-FC. (j) DSL-BC.

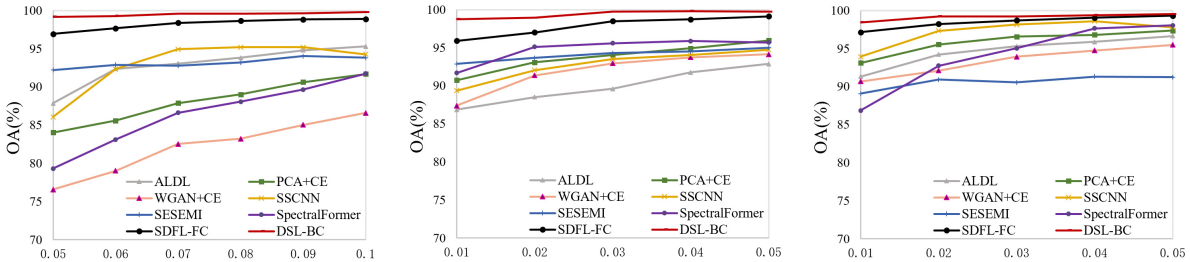


Fig. 5. Classification results of the ALDL, PCA+CE, WGAN+CE, SSCNN, SESEMI, SpectralFormer, SDFL-FC, and DSL-BC with different percentages of labeled image samples. (a) The Indian pines dataset. (b) The Salinas dataset. (c) The PaviaU dataset.

TABLE VIII
CLASSIFICATION RESULTS (OA%) WITH DIFFERENT COMPONENTS ON THE THREE DATASETS

Dataset	IR1	IR2	IR3	DSL-BC
Indian Pines	72.66	95.64	98.99	**99.17**
Salinas	89.49	98.19	98.28	**98.78**
PaviaU	90.95	98.32	98.34	**98.44**

The best results are highlighted in bold.

1) Without the GCN and superpixel constraint, the performance of the IR1 is lower than IR2, IR3, and DSL-BC for the three datasets.
2) For the three datasets, IR3 outperforms IR2. It indicates that, compared with the reconstruction error, the superpixel constraint can provide more helpful feedback for the GCN and prevent the salt-and-pepper problem.
3) Compared with IR1, IR2, and IR3, DSL-BC has superior performances because the DSL-BC integrates the DSL, the GCN, and the superpixel constraint into the joint learning framework.

E. Parameters Analysis

In the DSL-BC, five parameters λ_1, λ_2, the hidden node k_n of the GCN, the alternative parameter γ of the adjacency

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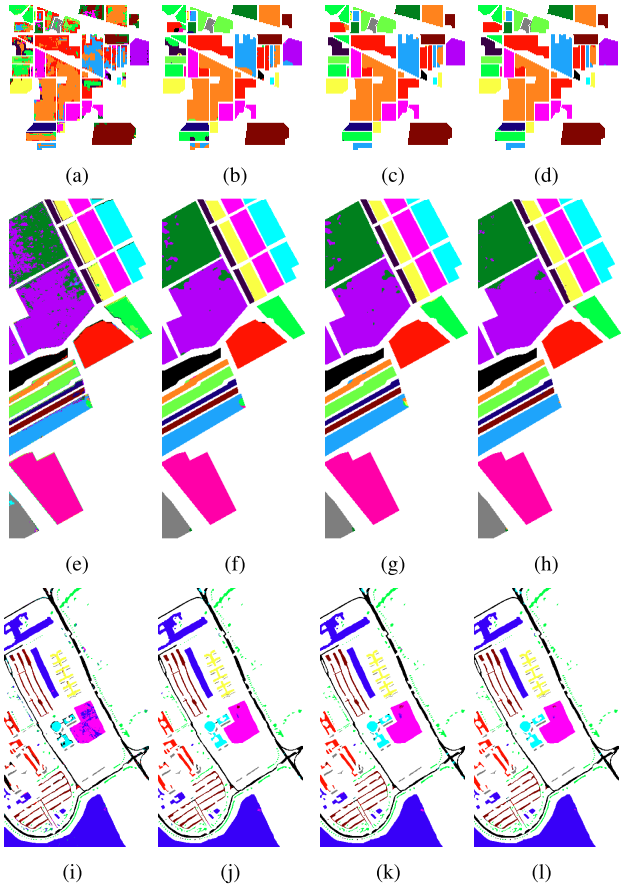


Fig. 6. Classification maps of different components of DSL-BC for the three datasets. (a)–(d) IR1, IR2, IR3, and DSL-BC for the Indian pines dataset, respectively. (e)–(h) IR1, IR2, IR3, and DSL-BC for the Salinas dataset, respectively. (i)–(l) IR1, IR2, IR3, and DSL-BC for the PaviaU dataset, respectively.

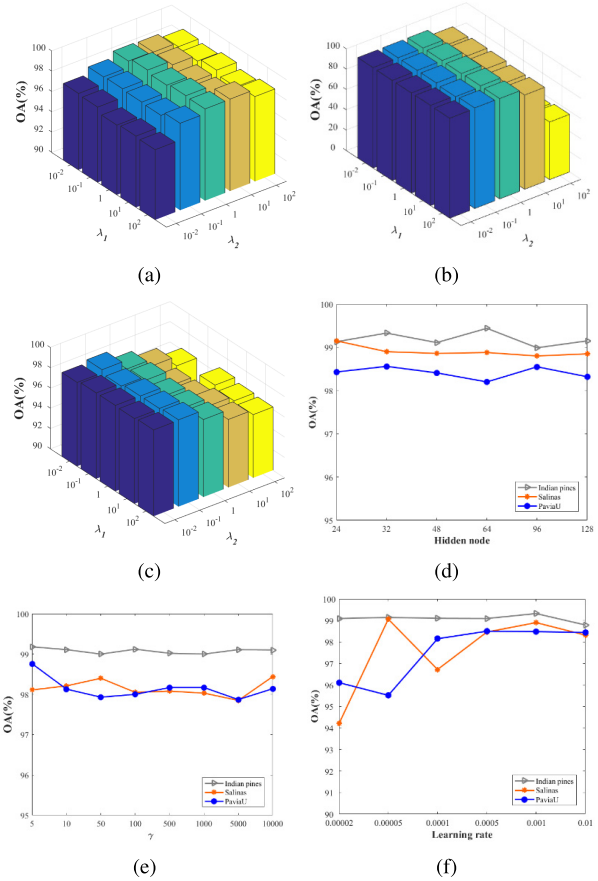


Fig. 7. Influences of the different parameter values on the HSI classification results. (a)–(c) Influences of λ_1 and λ_2 on the classification results for the Indian pines, Salinas, and PaviaU datasets, respectively. (d)–(f) Influences of different values of the hidden node k_n of the GCN, the alternative parameter γ of the adjacency matrix of the GCN, the learning rate on the classification results for the Indian pines, Salinas, and PaviaU datasets, respectively.

matrix of the GCN, the learning rate need to be tuned in three datasets. λ_1 and λ_2 correspond to the terms of reconstruction error and superpixel constraint, respectively. As shown in Fig. 7, we adjusted the parameters to find the best classification results. The structure of each dataset may be different. Therefore, for each dataset, the parameters that obtain the optimal results are different. The initial values of the parameters λ_1 and λ_2 are obtained based on the magnitude of different parts in the objective function. By varying values of five parameters, we test the performances. The optimal parameters should refer to the classification results of the testing data.

From Fig. 7, it is found that the value of λ_2 is larger than the value of λ_1 for the three datasets, which proves that the superpixel constraint has a more critical role in HSI classification. For the Indian pines and Salinas, the optimal λ_1 and λ_2 are close to 0.1 and 1.0. For the PaviaU dataset, the best λ_1 and λ_2 are to be 0.01 and 0.1.

For the Indian pines, the optimal parameters k_n, γ, and learning rate are close to 64, 5, and 0.001, respectively. For the Salinas dataset, the best image classification results are achieved when three parameters k_n, γ, and learning rate are close to 24, 10 000, 0.00005. For the PaviaU dataset, the optimal parameters k_n, γ, and learning rate are close to 32, 5, 0.0005. It is noted that the learning rate has more significant impacts than k_n and γ.

V. Conclusion

In this article, we have proposed the DSL-BC method to model the boundary consistency for the HSI classification. The DSL is obtained by CAE, which can capture the DSR. The GCN is further constructed by operating on the instance graphs to keep the boundary consistency. Moreover, the superpixel constraint prevents the salt-and-pepper problem and offers helpful feedback for SL and GCN. The experimental results on three challenging public HSI datasets demonstrate the advantages of DSL-BC over the related methods.

A global context is less explored in DSL-BC, indicating the margin of improving DSL-BC modeling. In future work, DSL will be worked with other state-of-the-art deep learning frameworks, for example, vision transformer, and also develop more advanced modules, to fully explore the global latent relationships of the HSI and further boost the HSI classification performance.

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